Caleb Penley

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Education

University of Maryland — College Park, MD
B.S., Mechanical Engineering, GPA: 3.6 — Expected December 2026

Skills

Software: SOLIDWORKS (CSWA), Siemens NX, MATLAB
Fabrication: 3D Printing (FDM/SLA/SLS), Lathe, Welding
Design / Analysis: FEA, 2D drawing creation, Tolerance stackup analysis, DFM, Root cause analysis

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Technical Experience

Mercedes-Benz U.S. International (MBUSI) — Vance, AL
Emissions/Testing Engineer, Co-op — May 2026-August 2026

- Operated a chassis dynamometer to run emissions test cycles and collect regulated pollutant data across six international certification programs.
- Produced audit-ready technical records, lab procedures, and knowledge transfer documentation ahead of a planned departmental shutdown.

Process Engineer, Co-op — August 2025-December 2025

- Diagnosed underperforming tools and processes through on-line observation and cross-functional troubleshooting, identifying root causes that led to two projects below.
- Designed a multi-spindle hand-start tool in SOLIDWORKS and Siemens NX, prototyped and iterated before manufacture, eliminating recurring safety incidents. In-house design removed supplier design fees entirely, cutting price from $11,800 to $3,600 (70%) and lead time from 10 to 2 weeks (80%).
- Restructured door-line process sequencing for a new vehicle model, reducing station utilization from 115% to 97%; negotiated design change feasibility with a 2-person R&D team in Germany over 4 biweekly Teams calls, and coordinated contractors to relocate overhead rails, lighting, and fans to accommodate the new CAN-tool layout.
- Tracked KPIs, milestones, and action items for a 16-person engineering team; managed cross-team project timelines using Excel Gantt charts.

Terrapin Works, Advanced Fabrication Lab — College Park, MD
Technician — January 2025-Present

- Operated FDM, SLA, and SLS printers across a range of polymer and composite materials, producing 1,271 prototypes; advised clients on DFM, tolerance expectations, and post-processing to ensure parts met functional requirements.

Formula SAE, Terps Racing (UMD) — College Park, MD
Vehicle Dynamics Engineer — August 2024-October 2025

- Researched and designed front and rear bellcrank suspension systems in SOLIDWORKS, connecting the anti-roll bar and pushrod to the spring/damper to reduce unsprung weight and improve handling over bumps and through corners.
- Used MATLAB to analyze suspension load data and kinematic model outputs, translating results into FEA boundary conditions and validating assumptions against physical test results.
- Reduced front and rear weight by 28.35% and 27.37% from the prior-year design using SOLIDWORKS FEA, sustaining 1,500 lbf front and 900 lbf rear pushrod loads at FOS 1.8 and 2.0, with fatigue life exceeding 11,900 cycles.
- Applied DFM principles for CNC machinability, cutting cost by 13.8% (front) and 9.9% (rear) from the prior year's design.

Additional Experience

Cru — College Park, MD
Treasurer, Community Group Leader, Outreach Team Leader — August 2022-Present

- Budgeted $15K+/semester for events and equipment serving 200+ students; led a bible study of 15+ and an outreach team with 75+ event participants.